

Normalization

Have we detected the right entities?

Every instance (a.k.a row) of this data shows the **involvement of a particular employee in a particular project**. It shows what the employee does for the project and how much they work and get paid.

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

Do you see any repetitions?!

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
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Repeating Groups of Attributes

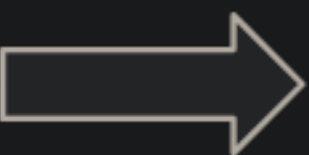
- Every project involves multiple employees.
- What would happen if we save all this data in a tabular format so that each cell of our table only includes a single value?

PROJECT NUMBER	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS	CHARGE/HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	\$ 84.50	23.8	\$ 2,011.10
		101	John G. News	Database Designer	\$105.00	19.4	\$ 2,037.00
		105	Alice K. Johnson *	Database Designer	\$105.00	35.7	\$ 3,748.50
		106	William Smithfield	Programmer	\$ 35.75	12.6	\$ 450.45
		102	David H. Senior	Systems Analyst	\$ 96.75	23.8	\$ 2,302.65
				Subtotal			\$10,549.70
18	Amber Wave	114	Annelise Jones	Applications Designer	\$ 48.10	24.6	\$ 1,183.26
		118	James J. Frommer	General Support	\$ 18.36	45.3	\$ 831.71
		104	Anne K. Ramoras *	Systems Analyst	\$ 96.75	32.4	\$ 3,134.70
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	44.0	\$ 2,021.80
				Subtotal			\$ 7,171.47
22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7	\$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	\$96.75	48.4	\$ 4,682.70
		113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6	\$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	\$26.87	22.0	\$ 591.14
		106	William Smithfield	Programmer	\$35.75	12.8	\$ 457.60
				Subtotal			\$13,660.10
25	Starflight	107	Maria D. Alonzo	Programmer	\$ 35.75	24.6	\$ 879.45
		115	Travis B. Bawangi	Systems Analyst	\$ 96.75	45.8	\$ 4,431.15
		101	John G. News *	Database Designer	\$105.00	56.3	\$ 5,911.50
		114	Annelise Jones	Applications Designer	\$ 48.10	33.1	\$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	\$ 96.75	23.6	\$ 2,283.30
		118	James J. Frommer	General Support	\$ 18.36	30.5	\$ 559.98
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	41.4	\$ 1,902.33
				Subtotal			\$17,559.82
				Total			\$48,941.09

Every row of this table represent a project. And a project seems to have repeating groups of attributes that show who are involved in the project.

(Identifier: PROJECT NUMBER)

PROJECT NUMBER	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS	CHARGE/ HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	\$ 84.50	23.8	\$ 2,011.10
		101	John G. News	Database Designer	\$105.00	19.4	\$ 2,037.00
		105	Alice K. Johnson *	Database Designer	\$105.00	35.7	\$ 3,748.50
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		101	John G. News *	Database Designer	\$105.00	56.3	\$ 5,911.50
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		108	Ralph B. Washington	Systems Analyst	\$ 96.75	23.6	\$ 2,283.30
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		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	41.4	\$ 1,902.33
		Subtotal					\$17,559.82
		Total					\$48,941.09



Repeating groups of attributes are removed, by turning that table into a new table in which every cell has a single value (not multiple values.)

Now every row of this table represents the involvement of a particular employee in a particular project. (Identifier: PROJ_NUM + EMP_NUM)

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
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A table is in **1 Normal Form (1NF)** if it does not contain attributes that have multiple values for a single instance of the entity.

This table is in **1 Normal Form (1NF)** and it has repetitions!

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
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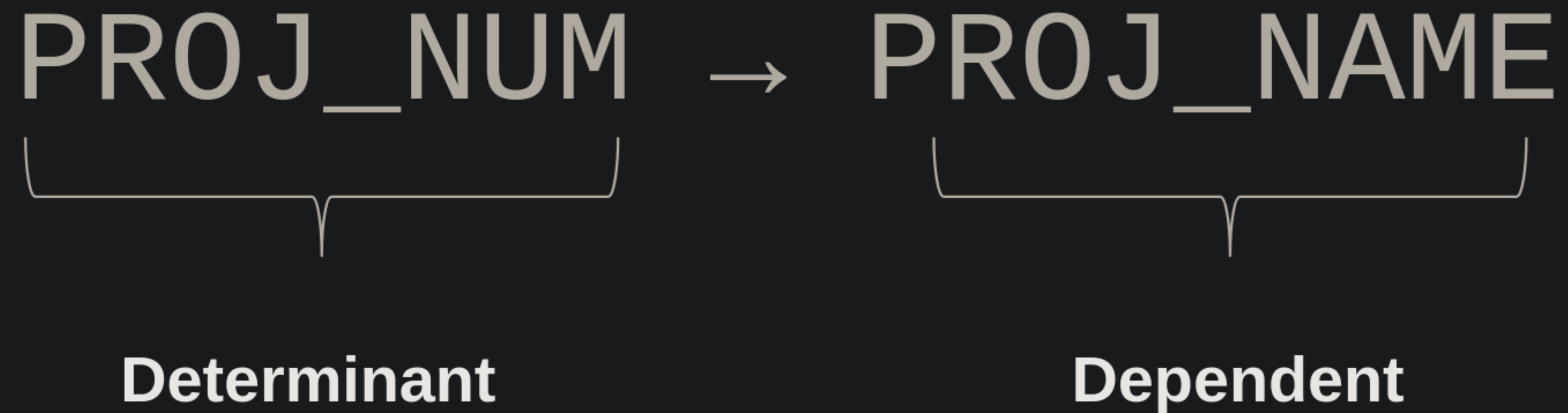
This table is in **1 Normal Form (1NF)** and it has repetitions!

PROJ_NUM	PROJ_NAME	EM
15	Evergreen	103
15	Evergreen	101
15	Evergreen	105
15	Evergreen	106
15	Evergreen	102
18	Amber Wave	114
18	Amber Wave	118
18	Amber Wave	104
18	Amber Wave	112
22	Rolling Tide	105
22	Rolling Tide	104
22	Rolling Tide	113
22	Rolling Tide	111
22	Rolling Tide	106
25	Starflight	107
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25	Starflight	101
25	Starflight	114
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William Smithfield	Programmer	35.75	12.8
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John G. News *	Database Designer	105.00	56.3
Annelise Jones	Applications Designer	48.10	33.1
Ralph B. Washington	Systems Analyst	96.75	23.6
James J. Frommer	General Support	18.36	30.5
Darlene M. Smithson	DSS Analyst	45.95	41.4

- Notice how every place PROJ_NUM is the same, PROJ_NAME also remains the same.
- In theory we say that PROJ_NUM **functionally defines** PROJ_NAME.
- Knowing a project's number, we can determine the project's name because only one name is associated with any given project number.

Functional Dependence



This table is in **1 Normal Form (1NF)** and it has repetitions!

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
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22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wahash	Clerical Support	26.87	22.0

Similarly, given an employee's number, we would know their name, job class and charge per hour:

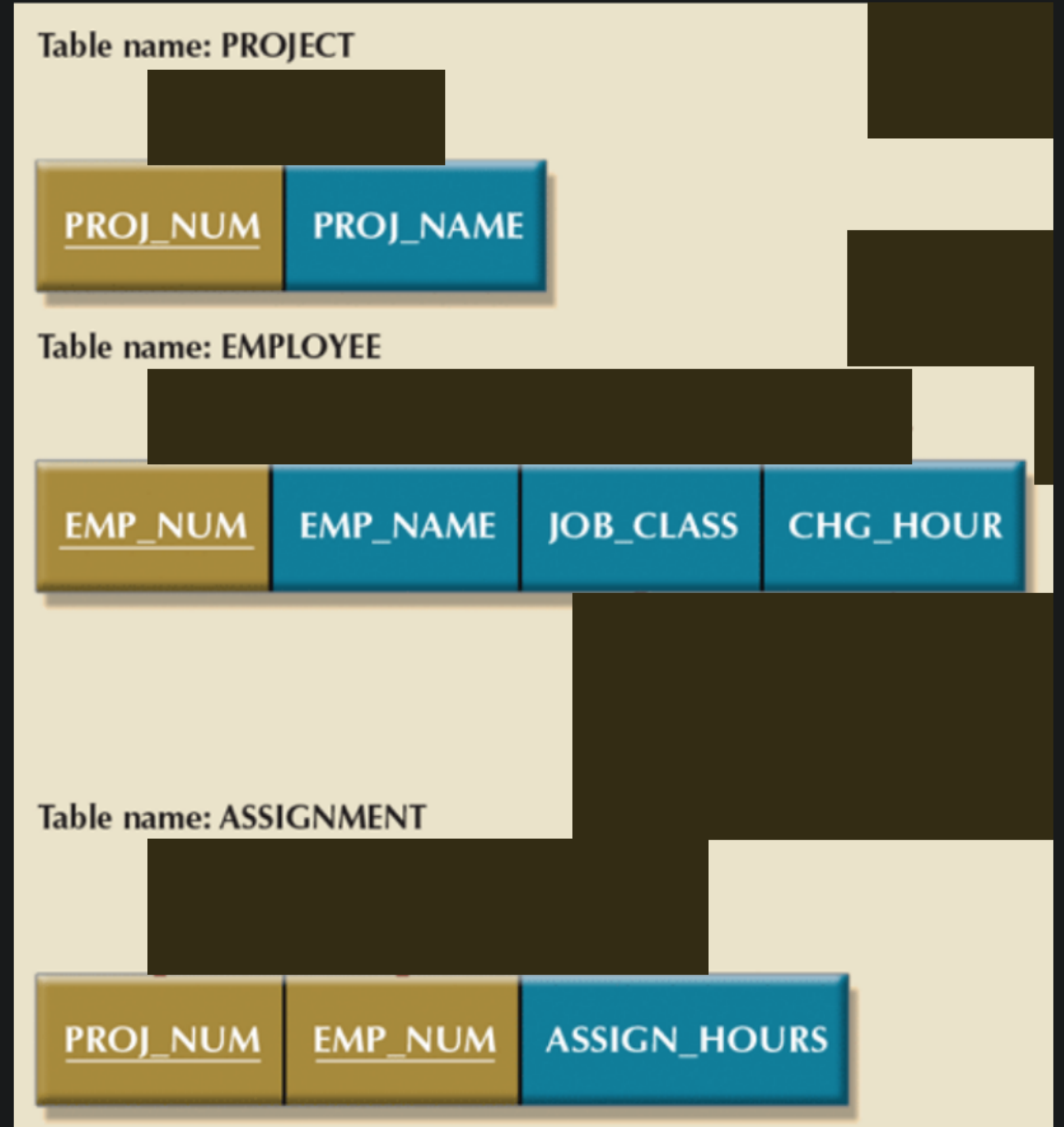
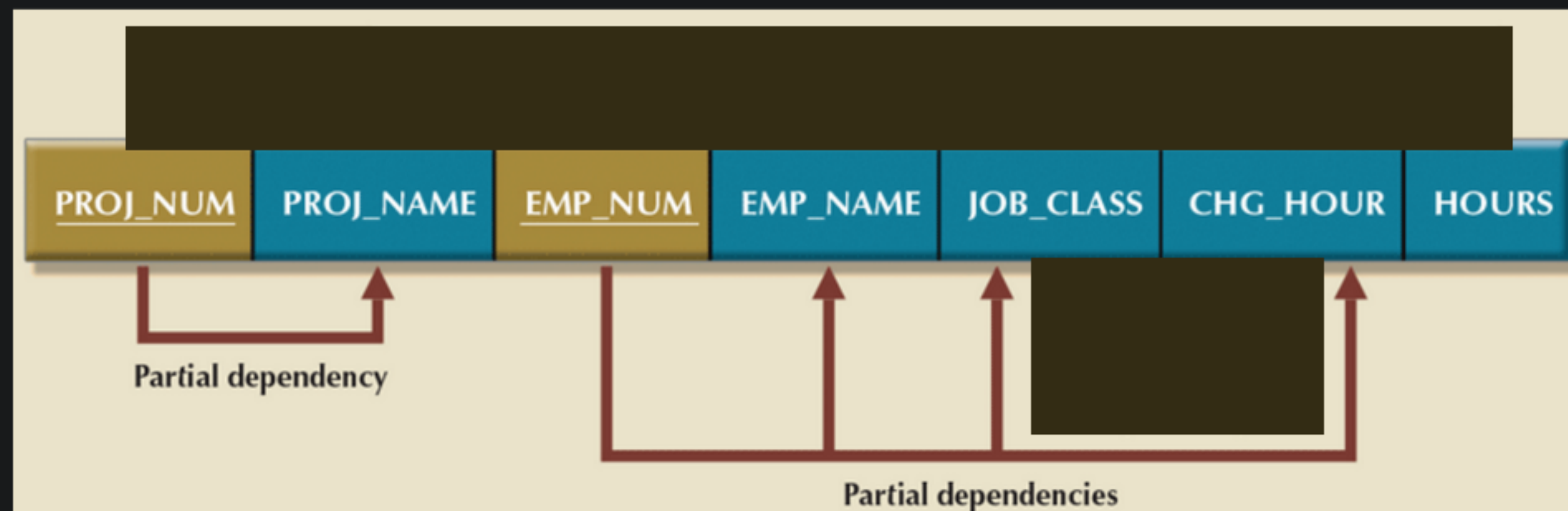
$EMP_NUM \rightarrow (EMP_NAME, JOB_CLASS, CHG_HOUR)$

Partial Dependency

- When non-identifier attributes are **dependent on only part of the identifier**.
 - Identifier: PROJ_NUM + EMP_NUM
 - PROJ_NUM $\rightarrow$ PROJ_NAME
 - EMP_NUM $\rightarrow$ (EMP_NAME, JOB_CLASS, CHG_HOUR)
- Partial dependence suggests that the **dependent attributes belong in another entity**.
- **Our goal** is to have entities containing attributes that are **dependent on the whole identifier** (not just a part of it).

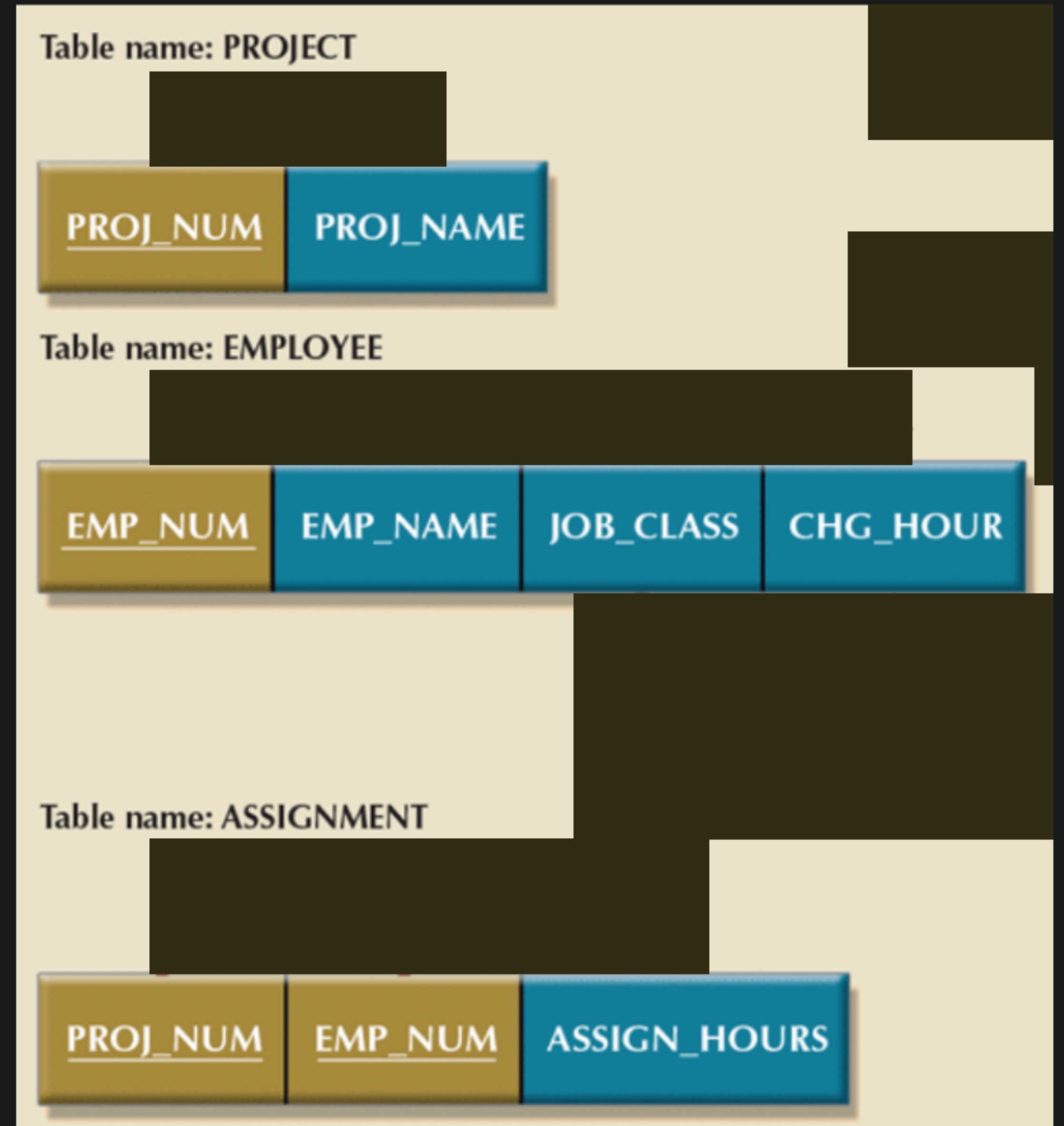
Removing Partial Dependencies

1. Create a new entity for every partial dependency.
2. Remove all dependent attributes from the **original entity**, but keep the rest of the attributes.



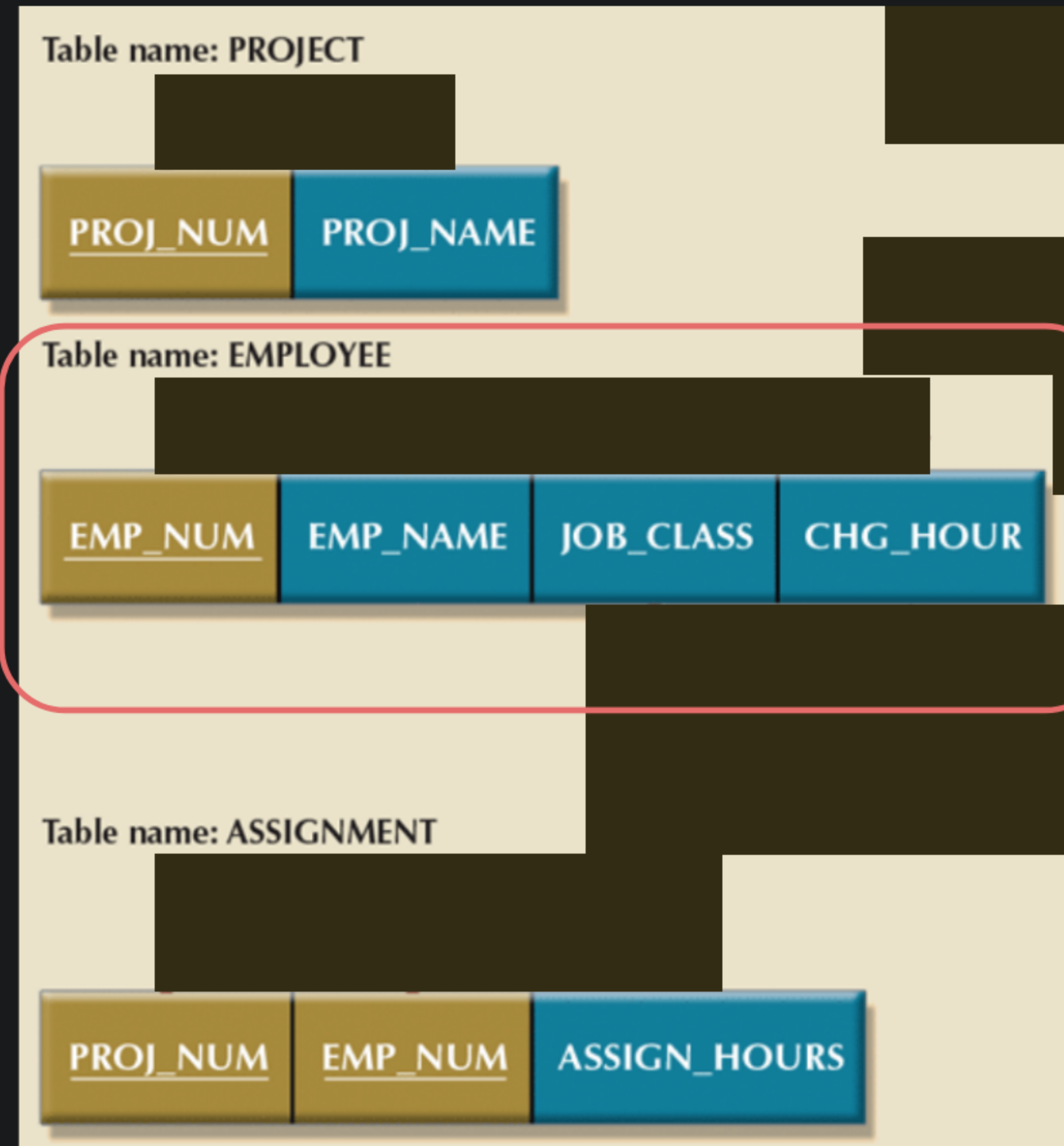
Second Normal Form (2NF)

These 3 entities are now in second normal form (2NF): They are in 1NF (**no repeating groups of attributes**) AND all attributes in an entity depend on the whole identifier (**no partial dependency**)



Second normal form (2NF)
requires that each entity is in
1NF and only has attributes
that depend on the whole
identifier (not a part of it).

The EMPLOYEE table is in 2NF and has repetition!



EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR
103	June E. Arbough	Elect. Engineer	84.50
101	John G. News	Database Designer	105.00
105	Alice K. Johnson *	Database Designer	105.00
106	William Smithfield	Programmer	35.75
102	David H. Senior	Systems Analyst	96.75
114	Annelise Jones	Applications Designer	48.10
118	James J. Frommer	General Support	18.36
104	Anne K. Ramoras *	Systems Analyst	96.75
112	Darlene M. Smithson	DSS Analyst	45.95

Notice how every place JOB_CLASS is the same, CHG_HOUR also remains the same.

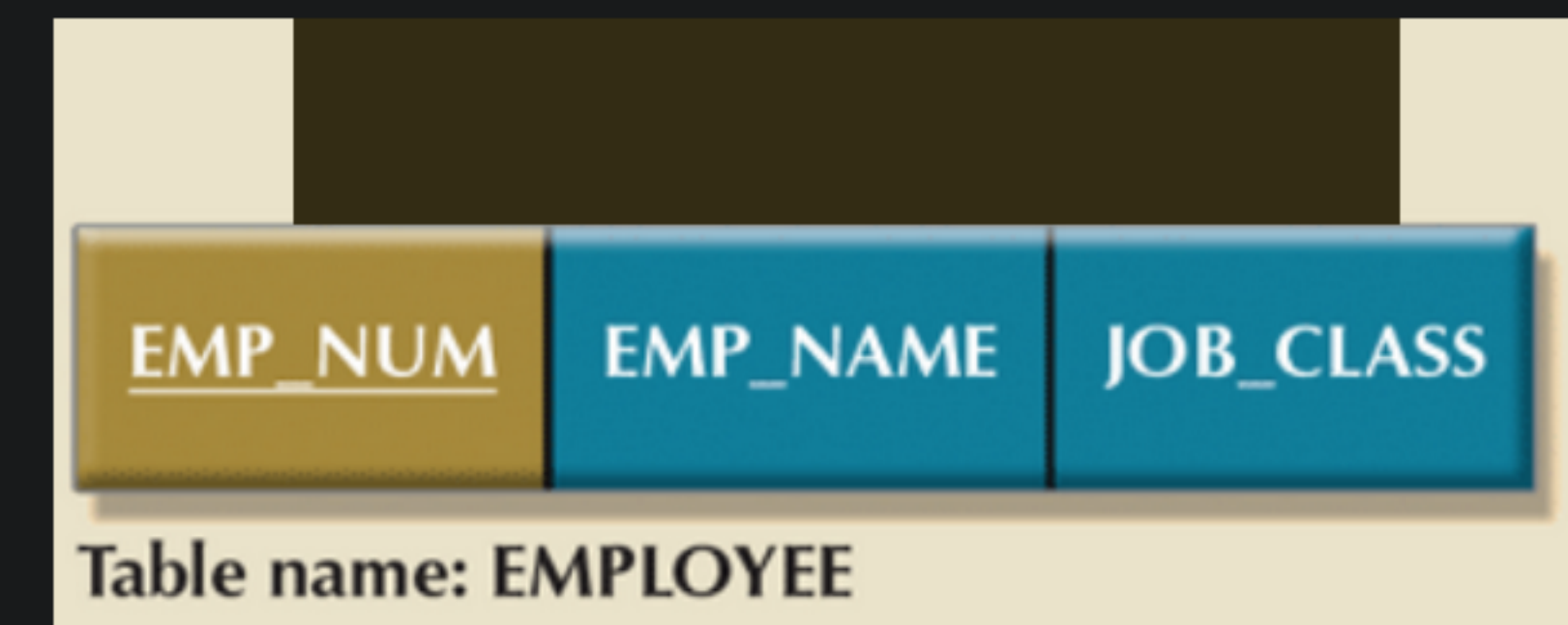
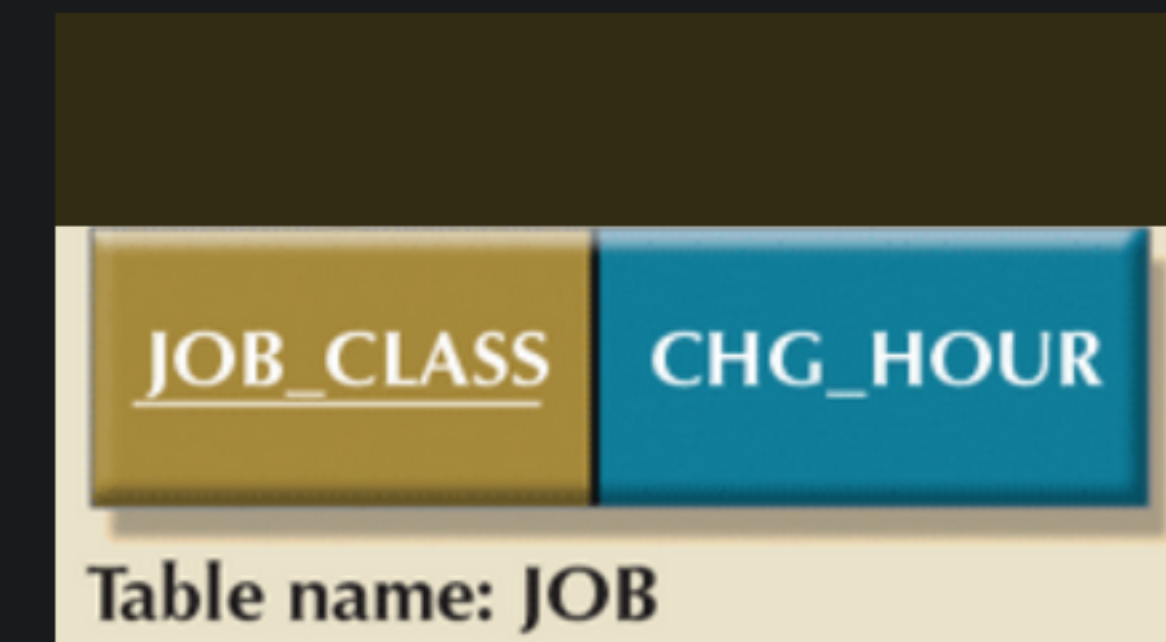
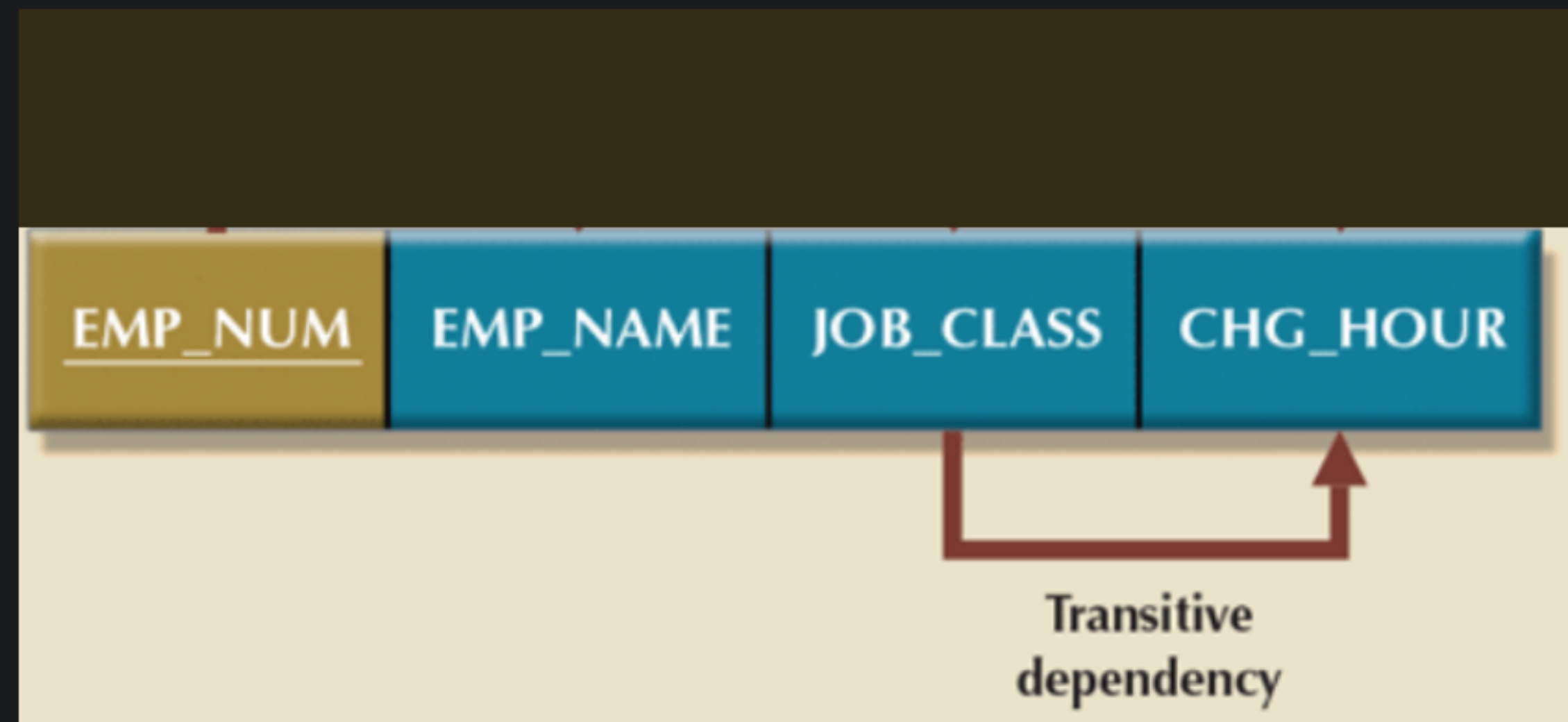
JOB_CLASS → CHG_HOUR

Transitive Dependency

- When an attribute **depends on a non-identifier attribute**.
 - Identifier: EMP_NUM
 - JOB_CLASS → CHG_HOUR
- Transitive dependency also suggests that the **dependent attributes belong in another entity**.

Removing Transitive Dependencies

1. Create a new entity for every transitive dependency.
2. Remove all dependent attributes from the **original entity**, but keep the rest of the attributes.



Third Normal Form (3NF)

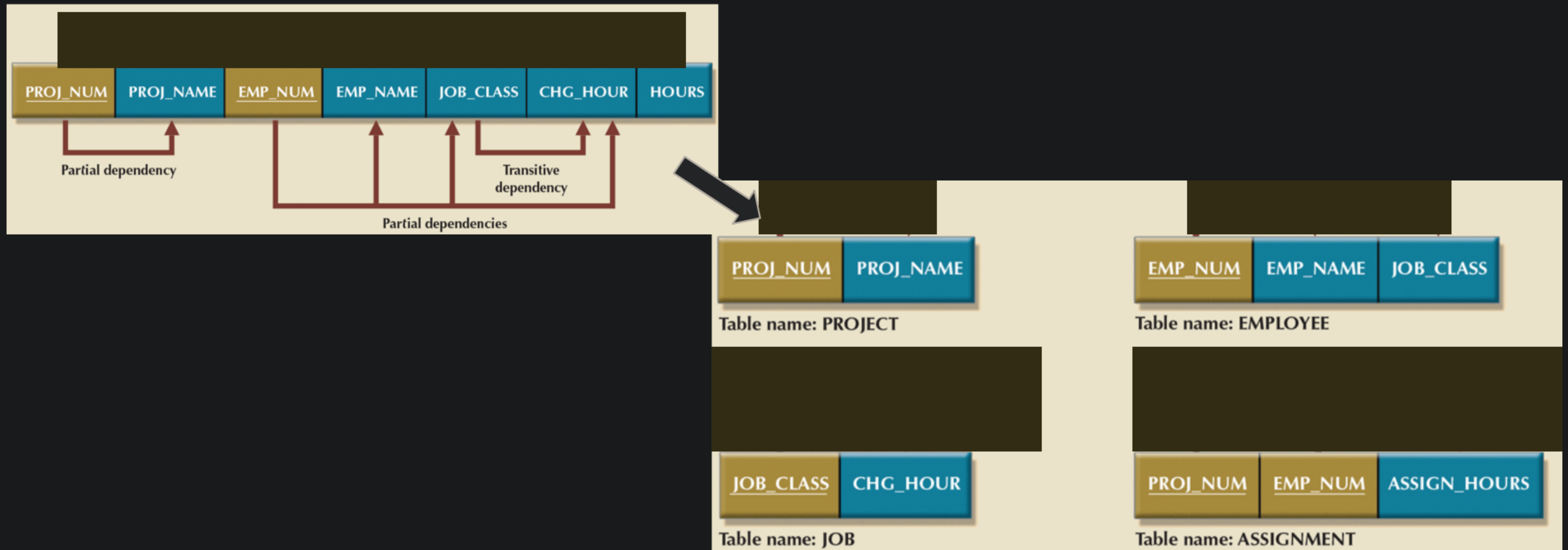
These 4 entities are now in third normal form (3NF): They are both in 1NF and 2NF (**no repeating groups of attributes, no partial dependency**) AND none of the attributes is dependent on a non-identifier attribute. (**no transitive dependency**)



Third normal form (3NF)
occurs when all entities are in
both 1NF and 2NF and when
none of attributes is
dependent on a non-identifier
attribute.

A Normalized Data Model

In practice we want data models where **all entities are in 3NF**.



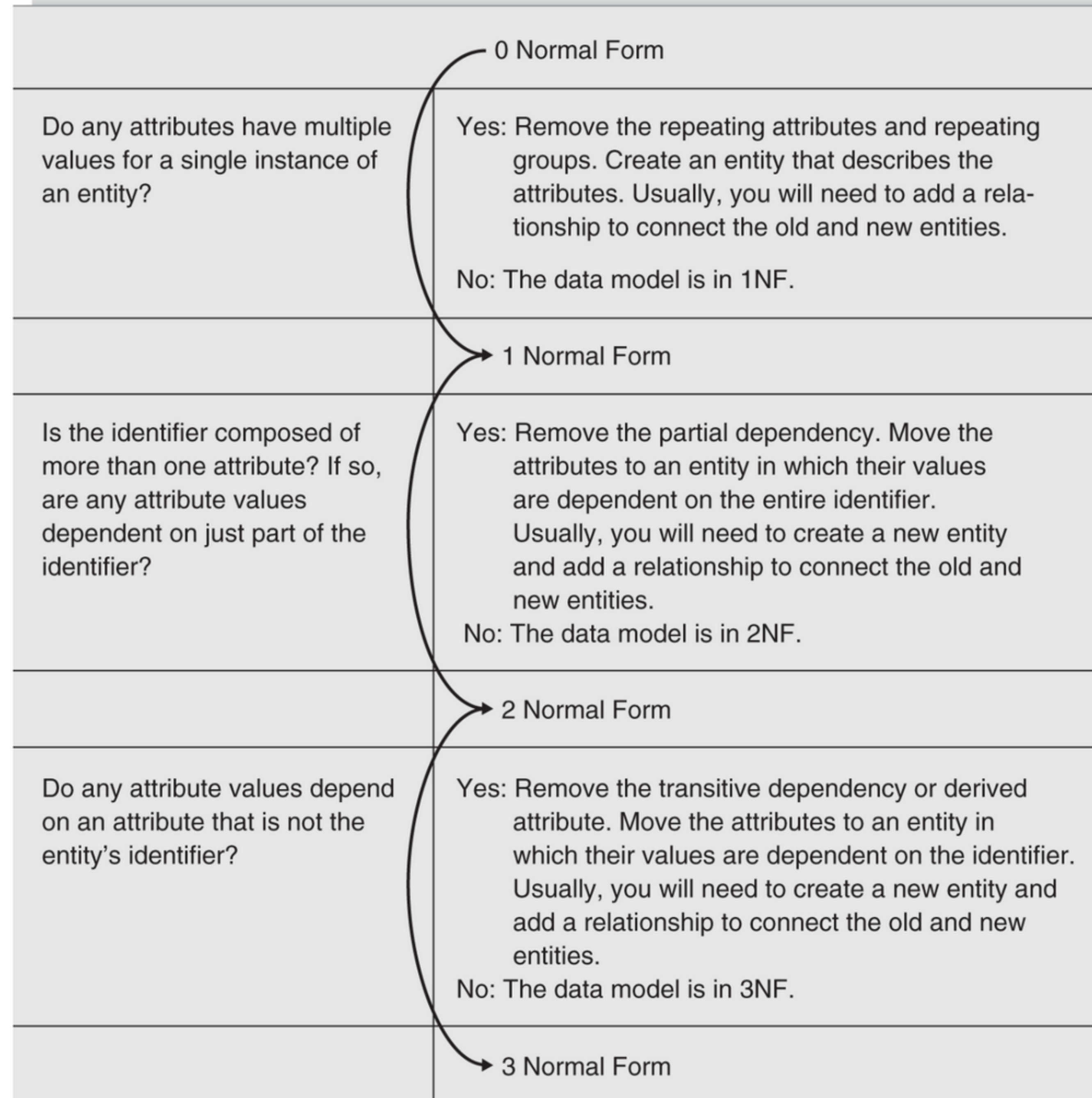
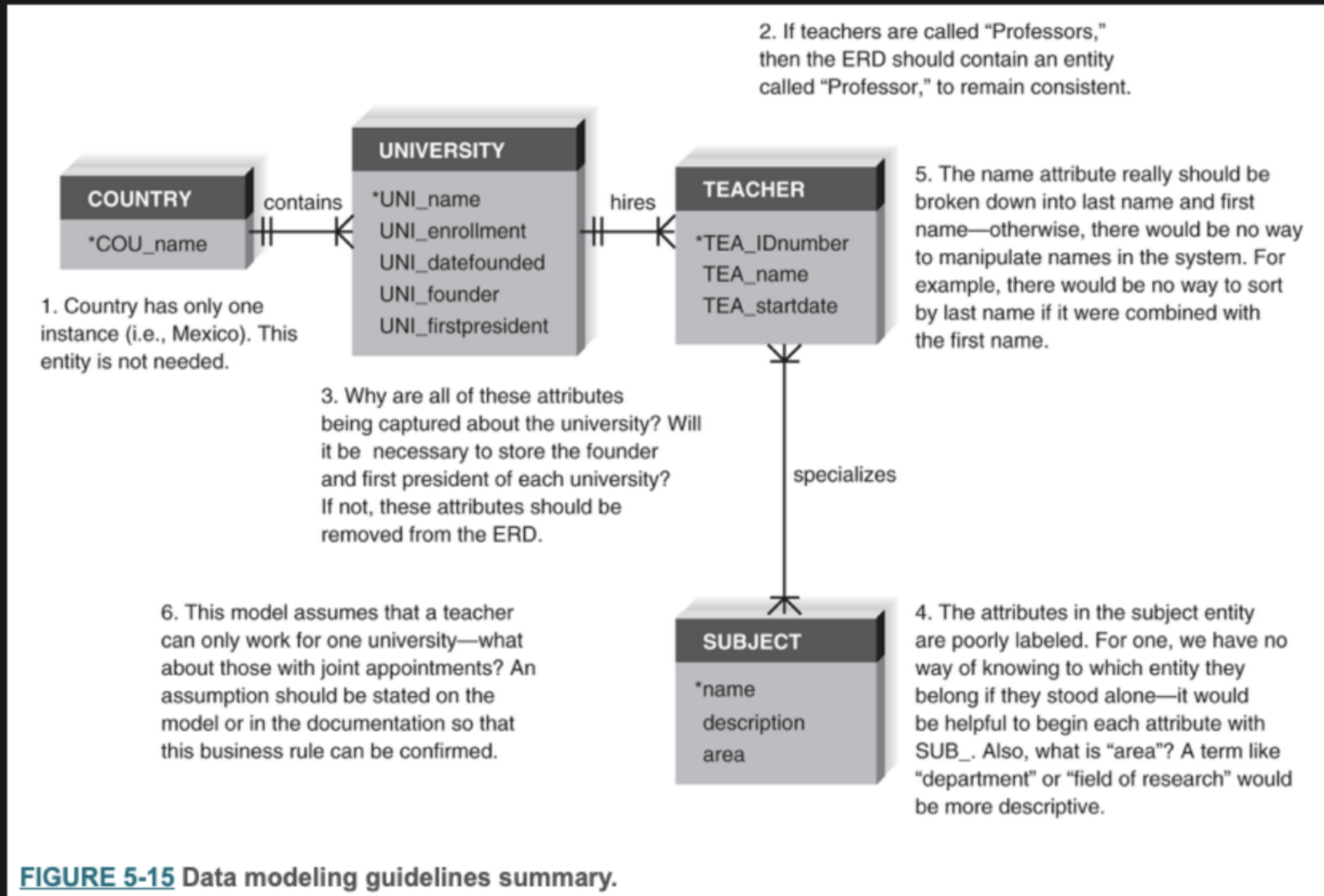


FIGURE 5A-1 Normalization steps.

Validating ERDs

- Normalization
- Design guidelines
- Balancing ERDs with DFDs

Design Guidelines



❏ Reading: Balancing ERDs with DFDs